

Open string spectrum :

• : trivial rep of $SO(24)$

Ground state: $|0; p\rangle$

Level	$\alpha' M^2$	state in $SO(24)$	little group	reps of little group
0	-1	$ 0\rangle$ • 1	$SO(28)$	• 1
1	0	$\alpha_{-1}^i 0\rangle$ $\square$ 24	$SO(24)$	$\square$ 24
2	+1	$\alpha_{-2}^i 0\rangle$ $\square$ 24 $\alpha_{-1}^i \alpha_{-1}^j 0\rangle$ $\square + \bullet$ (299) + 1	$SO(25)$	$\square$ 324
3	+2	$\alpha_{-3}^i 0\rangle$ $\square$ 24 $\alpha_{-2}^i \alpha_{-1}^j 0\rangle$ $\square + \square + \bullet$ 276 299 1 $\alpha_{-1}^i \alpha_{-1}^j \alpha_{-1}^k 0\rangle$ $\square + \square$ (2576) 24	$SO(28)$	$\square$ 2900 + $\square$ 300
4	+3	$\alpha_{-4}^i 0\rangle$ $\square$ 24 $\alpha_{-3}^i \alpha_{-1}^j 0\rangle$ $\square + \square + \bullet$ 276 299 + 1 $\alpha_{-2}^i \alpha_{-1}^j \alpha_{-1}^k 0\rangle$ $2 \times \square + \square + \square$ $2 \times 24 + 2576 + 4576$ $\alpha_{-1}^i \alpha_{-1}^j \alpha_{-1}^k \alpha_{-1}^l 0\rangle$ $\square + \square + \bullet$ 17250 + 299 + 1	$SO(28)$	$\square$ 20150 + $\square$ 5175 + $\square$ 324 + $\bullet$ 1

* Expanding on my analysis on the notes:

Level 3: $\alpha_{-3}^i A_i |0\rangle + \alpha_{-2}^i \alpha_{-1}^j B_{ij} |0\rangle$
+ $\alpha_{-1}^i \alpha_{-1}^j \alpha_{-1}^k C_{ijk} |0\rangle$

Any 2-tensor can be broken into the

following:

$$B_{ij} = \underbrace{B_{[ij]}}_{\square} + \underbrace{(B_{(ij)} - \delta^{ik} B_{ke})}_{\square} + \underbrace{\delta_{ij} \frac{\text{tr}(B)}{D-2}}_{\underline{1}}$$

* at level $n \geq 2$ this means that

* $\alpha_{-1}^i \alpha_{-1}^j B_{[ij]} = 0$

Since $\alpha_{-1}^i \alpha_{-1}^j$ are "identical" we cannot have them be antisymmetric.

$$\begin{aligned} \frac{1}{2} (\alpha_{-1}^i \alpha_{-1}^j) (B_{[ij]} - B_{[ji]}) &= \frac{1}{2} (\alpha_{-1}^i \alpha_{-1}^j B_{[ij]} - \alpha_{-1}^j \alpha_{-1}^i B_{[ji]}) \\ &= \frac{1}{2} (\alpha_{-1}^i \alpha_{-1}^j B_{[ij]} - \alpha_{-1}^j \alpha_{-1}^i B_{[ji]}) \end{aligned}$$

use $[\alpha_{-1}^i, \alpha_{-1}^j] = 0$

Relabeling $i \leftrightarrow j$ on the second term yields $\alpha_{-1}^i \alpha_{-1}^j B_{[ij]} = 0$

So this is NOT a rep of $so(D-2)$

* Now, at level 3: we have $\alpha_{-2}^i \alpha_{-1}^j B_{[ij]} |0\rangle$
we still have the decomp of $B_{[ij]}$ but

$$\begin{aligned} \alpha_{-2}^i \alpha_{-1}^j B_{[ij]} &= \frac{1}{2} (\alpha_{-2}^i \alpha_{-1}^j B_{[ij]} - \alpha_{-1}^i \alpha_{-2}^j B_{[ji]}) \\ &= \frac{1}{2} \alpha_{-2}^{[i} \alpha_{-1}^{j]} B_{[ij]} \end{aligned}$$

which is non-zero. Indeed, Plugging the decomp $(B \equiv \text{tr}(B))$

$$B_{ij} = B_{[ij]} + (B_{(ij)} - B) + \left(\frac{B}{D-2}\right) \delta_{ij}$$

into $\alpha_{-2}^i \alpha_{-1}^j B_{[ij]} |0\rangle$ we obtain

$$\alpha_{-2}^i \alpha_{-1}^j B_{[ij]} = \underbrace{(\alpha_{-2}^i \alpha_{-1}^j B_{[ij]})}_{\square} + \underbrace{\alpha_{-2}^i \alpha_{-1}^j (B_{(ij)} - B)}_{\square} + \underbrace{\alpha_{-2}^i \alpha_{-1}^j \frac{B}{D-2} \delta_{ij}}_{\underline{1}}$$

→ In the main notes I left out $\square$

as I assumed (wrongly) that B_{ij} was symmetric.

So far we have

$$\square + \frac{(D-2)(D-3)}{2} \square + \frac{(D-1)(D-2)}{2} \bullet = 1$$

(D-2)

* Must look at C_{ijk} :

(c) Note C_{ijk} cannot be anti-symmetric in any of its indices. Thus

$$\square \otimes \square \otimes \square = \square \times (\square + \square + \bullet)$$

$$= \square \otimes \square \otimes \square + \square \otimes \square \otimes \square + \square$$

Note: I have manually added $(+ \bullet)$ to the above as this is the decomp of a full rank 2 $B_{ij} \sim \square \otimes \square$

Now, $\square$ and $\square$ are anti-symmetric and will vanish leaving:

$$C_{ijk} \sim \square \otimes \square + \square$$

• To understand the meaning of $\square$ is simple. We take the "vector trace" of the rank 3 tensor

$$C_{ijk} \rightarrow \delta^{ij} C_{ijk} \sim V_k$$

which transforms as a vector under $SO(D-2)$

* The full level 3 massless spectrum is therefore

$$\square + \frac{(D-2)(D-3)}{2} \square + \frac{(D-1)(D-2)}{2} \square + \underline{1} + \frac{(D-2)(D-1)}{6} \square + \underline{D-2}$$

Included Included

Major NOTE: 1 and D-2 are already part

of II and III respectively.
 II does not remove the traceless part and
 likewise for III so they are NOT
INCLUDED:

$$\begin{aligned}
 &= \frac{(D-2)6}{6} [6 + 3D - 9 + 3D - 3 + D^2 - D] \\
 &= \frac{D-2}{6} (6D - 6 - D + D^2) \\
 &= \frac{D-2}{6} (D^2 + 5D - 6) = \frac{(D-2)(D+6)(D-1)}{6}
 \end{aligned}$$

$$\therefore \frac{(D-2)(D+6)(D-1)}{6}$$

We find for $D=26$ this is 3200 ✓.

* So, Basically problem was not including A in the sum.